

Minecraft Crafting Reference

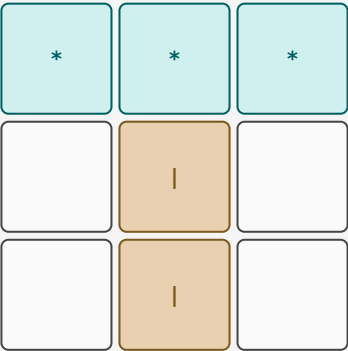
A complete visual guide to placing items on the 3×3 crafting table. By Walusimbi Leon (SGSS) · Public Domain

- Diamond
- Gold
- Iron
- Redstone
- Planks
- Cobblestone
- Obsidian
- Stick
- Special Item
- Wood/Log

1. Tools

Diamond Pickaxe

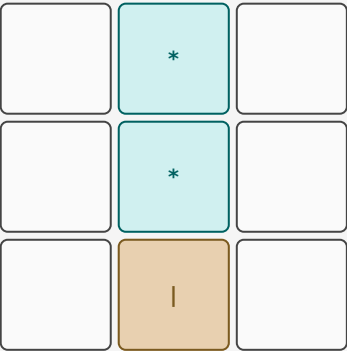
Yield: 1 pickaxe



Tip: First diamond item you should make. Swap diamond with iron/wood for lower tiers.

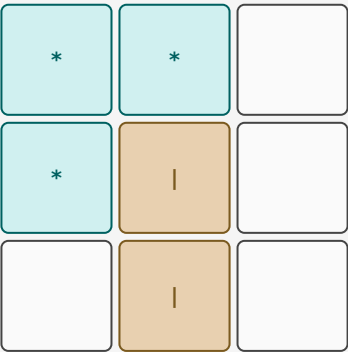
Diamond Sword

Yield: 1 sword



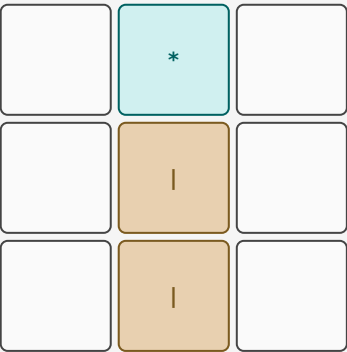
Diamond Axe

Yield: 1 axe



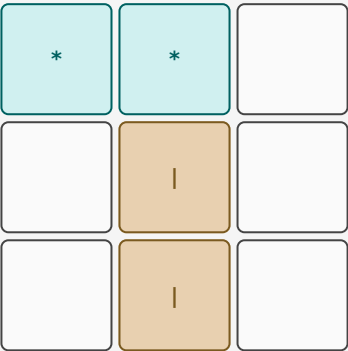
Diamond Shovel

Yield: 1 shovel



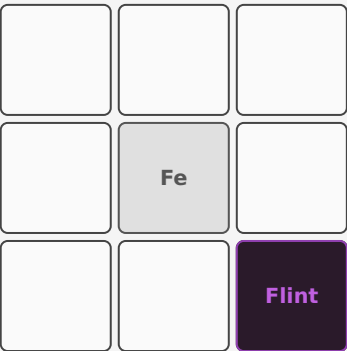
Diamond Hoe

Yield: 1 hoe



Flint and Steel

Yield: 1 flint & steel



Shears

Yield: 1 shears

	Fe	
		Fe

Iron on one diagonal. Collect leaves, get wool without killing sheep, mine cobwebs.

Fishing Rod

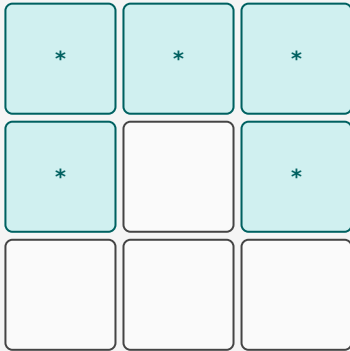
Yield: 1 fishing rod

		String

2. Armor

Diamond Helmet

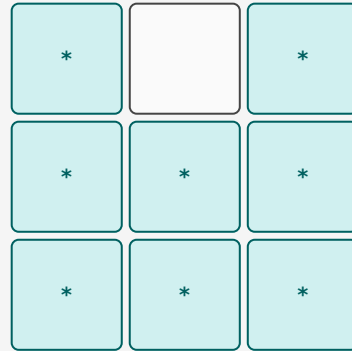
Yield: 1 helmet | 5 diamonds



Note: Replace diamond with iron, gold, or leather for lower tiers.

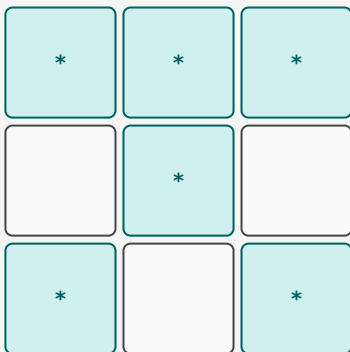
Diamond Chestplate

Yield: 1 chestplate | 8 diamonds



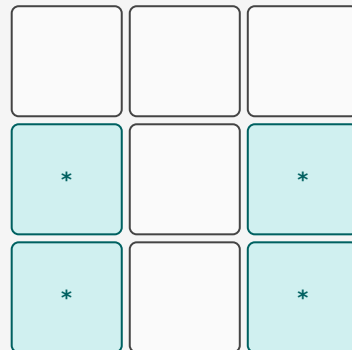
Diamond Leggings

Yield: 1 leggings | 7 diamonds



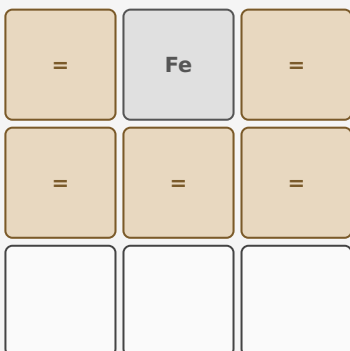
Diamond Boots

Yield: 1 boots | 4 diamonds



Shield

Yield: 1 shield

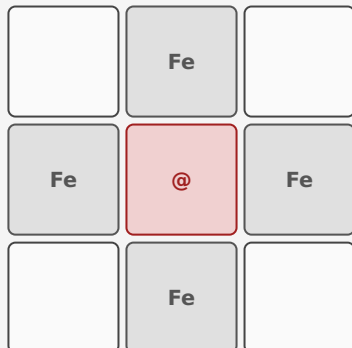


Blocks 100% damage when raised.
One of the best early-game items.

3. Utility & Navigation

Compass

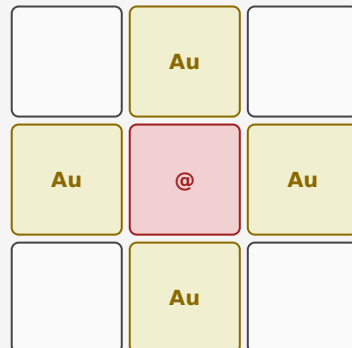
Yield: 1 compass



Points to world spawn. Spins randomly in Nether/End.

Clock

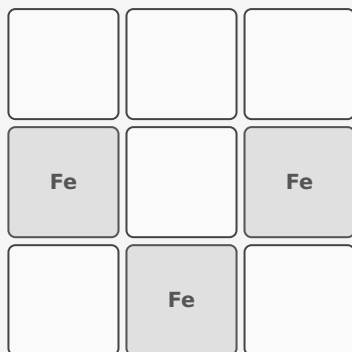
Yield: 1 clock



Tells time (sun/moon). Works underground. Gimmicky but cool.

Bucket

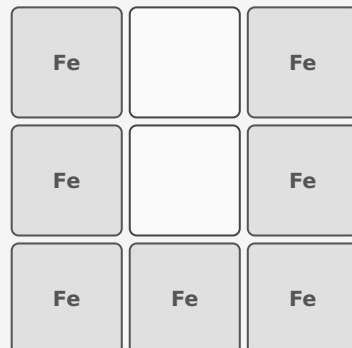
Yield: 1 bucket



Carry water, lava, or milk. Water bucket negates fall damage if placed under you.

Cauldron

Yield: 1 cauldron

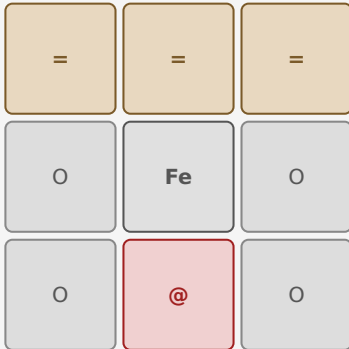


Holds water, removes dye from leather armor, holds lava.

4. Machines & Contraptions

Piston

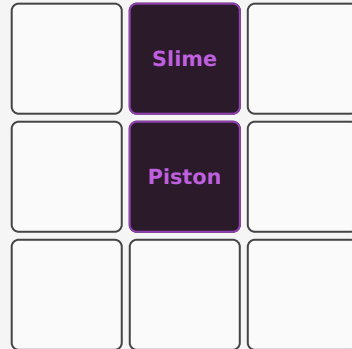
Yield: 1 piston



Pushes blocks. Combine with sticky piston (slime ball) to pull too. Foundation of redstone contraptions.

Sticky Piston

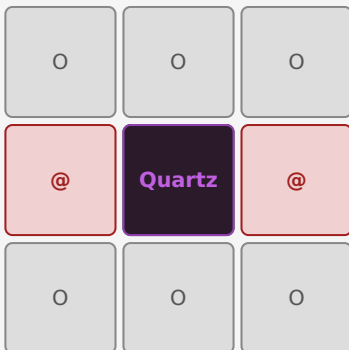
Yield: 1 sticky piston



Place a slime ball on top of a piston. Pulls blocks back too.

Observer

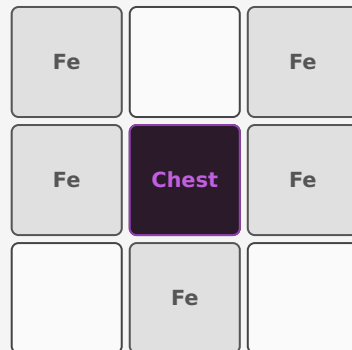
Yield: 1 observer



Emits redstone pulse when the block it faces changes. Automate farms.

Hopper

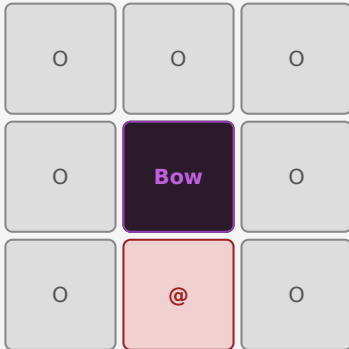
Yield: 1 hopper



Collects items above it, feeds into inventory below. Essential for auto-farms and auto-smelters.

Dispenser

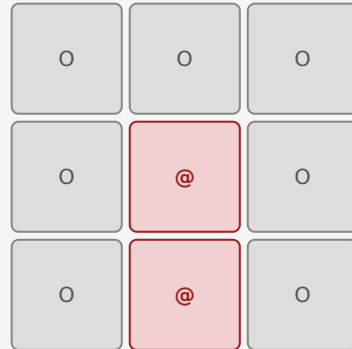
Yield: 1 dispenser



Shoots items when activated.
Place arrows → auto-bow, water
bucket → places water, bonemeal
→ fertilizes.

Dropper

Yield: 1 dropper

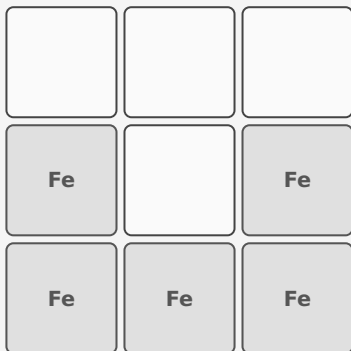


Drops items out. Unlike dispenser,
it just spits items — doesn't
activate them.

5. Transportation

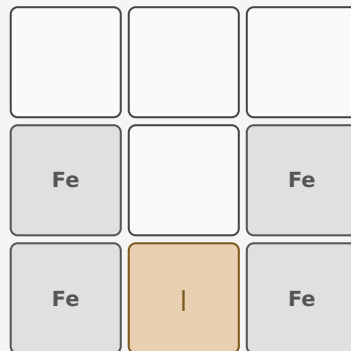
Minecart

Yield: 1 minecart



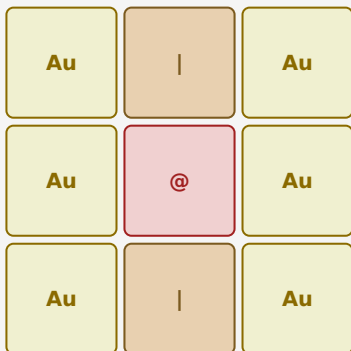
Rail

Yield: 16 rails



Powered Rail

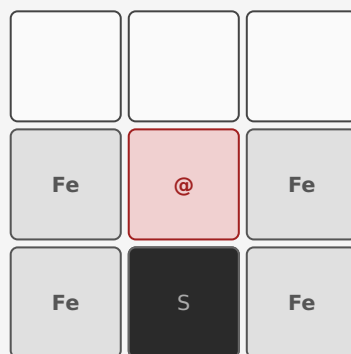
Yield: 6 rails



Accelerates minecarts. Needs redstone power to work. Place every ~32 blocks for speed.

Detector Rail

Yield: 6 rails

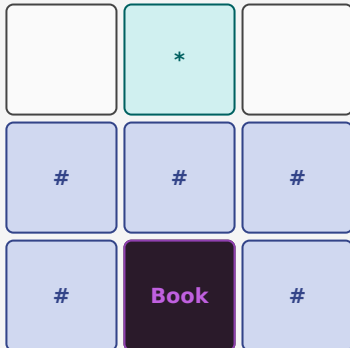


Stone pressure plate in center-bottom. Emits redstone when minecart passes over.

6. Enchanting & Upgrades

Enchanting Table

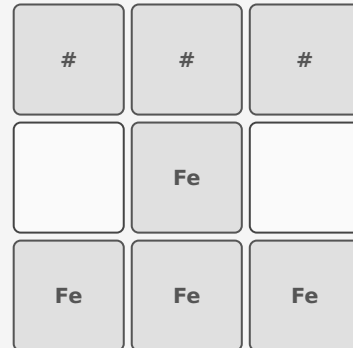
Yield: 1 table



Book: 3 paper + 1 leather. Make this as soon as you have 2 diamonds + 4 obsidian.

Anvil

Yield: 1 anvil

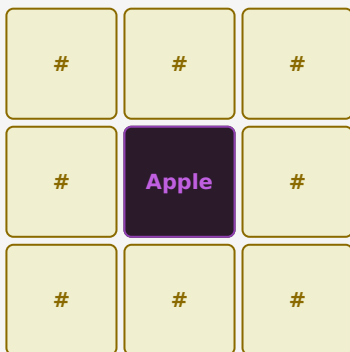


Top row = 3 iron blocks (27 ingots). Bottom row = 4 ingots. Repair items, combine enchantments.

7. Consumables & Food

Golden Apple

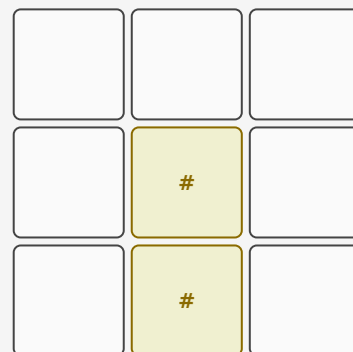
Yield: 1 golden apple



8 gold blocks (72 ingots!) + 1 apple. Absorption II + Regen II. Lifesaver for Nether/raids/final fights.

Golden Carrot

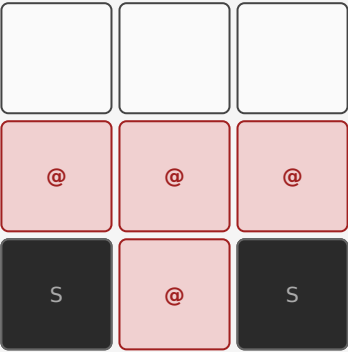
Yield: 1 golden carrot



8. Basic Redstone Components

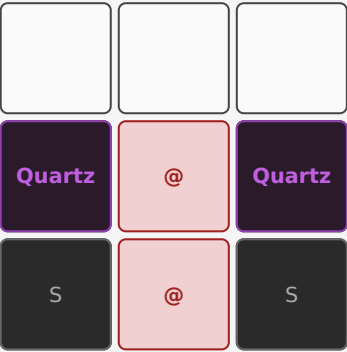
Redstone Repeater

Yield: 1 repeater



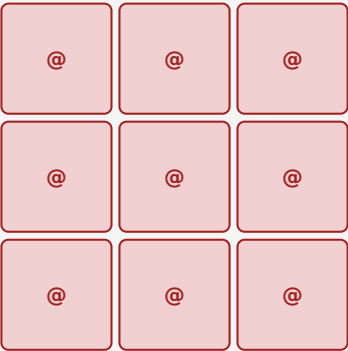
Redstone Comparator

Yield: 1 comparator



Redstone Block

Yield: 1 redstone block



Permanent redstone power source.
Also powers beacons.

9. Storage & Building

Chest

Yield: 1 chest

=	=	=
=		=
=	=	=

Furnace

Yield: 1 furnace

o	o	o
o		o
o	o	o

Crafting Table

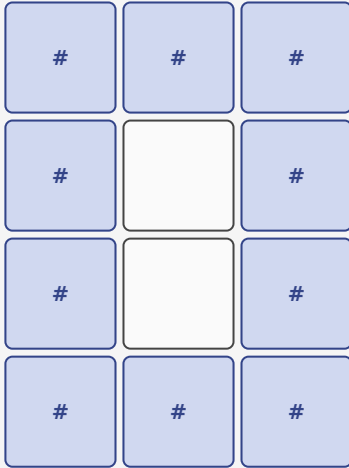
Yield: 1 crafting table

=	=	
=	=	

10. Nether Portal

Nether Portal (not craftable)

Requires: 10-14 obsidian + flint & steel

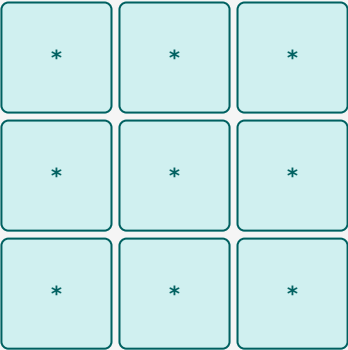


Build a vertical frame of obsidian (min 4×5, max 23×23). Light inside with flint & steel. Get obsidian by pouring water over lava source blocks.

11. Block Crafting (Compression)

Block of Diamond

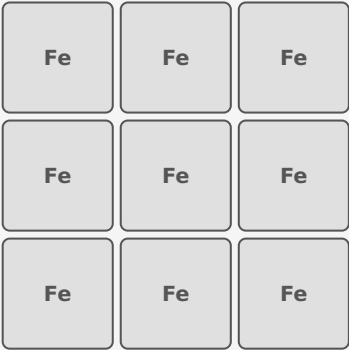
Yield: 1 block



9 diamonds → 1 block. Compact storage. Use for beacon bases.

Block of Iron

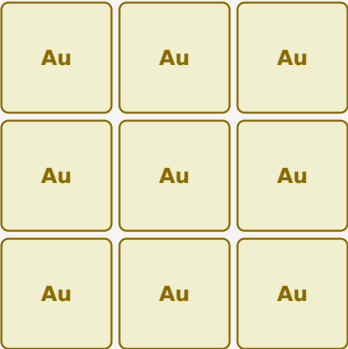
Yield: 1 block



9 iron → 1 block. Use for anvil top row (3 blocks), beacon bases, iron golems.

Block of Gold

Yield: 1 block



9 gold → 1 block. Used for golden apple (8), beacon bases, piglin bartering.

12. Quick Reference: Crafting Positions

Item	Grid Pattern (top→bottom)	Materials
Pickaxe	XXX / _S_ / _S_	3 material + 2 sticks

Axe	XX_ / XS_ / _S_	3 material + 2 sticks
Sword	_X_ / _X_ / _S_	2 material + 1 stick
Shovel	_X_ / _S_ / _S_	1 material + 2 sticks
Hoe	XX_ / _S_ / _S_	2 material + 2 sticks
Helmet	XXX / X_X	5 material
Chestplate	X_X / XXX / XXX	8 material
Leggings	XXX / _X_ / X_X	7 material
Boots	X_X / X_X	4 material
Block (compressed)	XXX / XXX / XXX	9 ingots/gems
Shield	PFeP / PPP	6 planks + 1 iron
Bucket	Fe_Fe / _Fe_	3 iron ingots
Compass	_Fe_ / Fe@Fe / _Fe_	4 iron + 1 redstone
Piston	PPP / CFeC / C@C	3 planks, 4 cobble, 1 iron, 1 redstone

Legend: X = material (diamond/iron/gold/stone/wood) · S = stick · P = plank · C = cobblestone · Fe = iron · @ = redstone · # = block · Au = gold · _ = empty